RAMDEOBABA UNIVERSITY, NAGPUR

Department of Electronics Engineering

CAPSTONE PROJECT PROPOSAL

Course: Deep Learning Lab | Academic Session: 2026–27 | Semester: V

1. Problem Statement and Motivation

People who need assistance while walking may face curbs, steps, poles, vehicles, people and blocked pathways. A normal camera feed does not directly indicate which part of the scene can be used for walking. This project will use semantic segmentation to identify four useful classes: walkable area, non-walkable area, dynamic obstacles and structural obstacles. Cityscapes will be used for the main urban-scene training, followed by fine-tuning on a small campus dataset containing accessibility-related cases. Since the final system is intended to run on a Jetson Orin Nano, the model will be selected with both accuracy and real-time speed in mind. The final prototype will show a live camera feed with a walkable-region overlay and obstacle highlighting. It is intended as an assistive research prototype and not as a certified mobility or safety device.

Project Information

Project Title				
Team Member 1	**Name:**	**Divyansh Gupta**	**Roll No.:**	**22**
Team Member 2	**Name:**	**Arnav Pangul**	**Roll No.:**	**14**
Team Member 3	**Name:**	**Gauri Thakre**	**Roll No.:**	**23**
Team Member 4	**Name:**	**Dhruv Runwal**	**Roll No.:**	**21**

2. Project Objectives

1. Convert relevant Cityscapes labels into four project classes: Walkable, Non-walkable, Dynamic Obstacle and Structural Obstacle.
2. Train and compare two lightweight semantic-segmentation models, with BiSeNetV2 as the main model and Fast-SCNN as the baseline.
3. Collect and annotate approximately 300–500 campus/public-area images covering curbs, steps, poles, vehicles, people and blocked pathways.
4. Fine-tune the selected model on the local dataset and check whether it improves performance on campus scenes.
5. Deploy the selected model on Jetson Orin Nano and target at least 20 FPS end-to-end while measuring latency, memory and power.

3. Dataset

Dataset Name	Cityscapes + Local Campus Dataset
Dataset Source	Official Cityscapes dataset; Manual Collection of Images from Locality.
Dataset Size	5,000 finely & 20,000 coarsely annotated images, 300–500 self-collected
Type of Data	Annotated RGB Images

No. of Classes / Outputs	4 classes: Walkable, Non-walkable, Dynamic Obstacle, Structural Obstacle

4. Proposed Methodology

Camera → Resize & Normalize → Fast-SCNN/BiSeNetV2 segmentation → 4-class Segmentation Mask → Walkable Region/Obstacle Map → Walking Zone Analysis → Visual/Audio Alert

Model choice: BiSeNetV2 will be the main model because it is designed for real-time semantic segmentation and keeps spatial details while also learning higher-level scene information. Fast-SCNN will be used as a lighter baseline so that we can compare the accuracy and speed trade-off. The final model will be selected only after testing both models on our data and on the Jetson Orin Nano.

Four-class output: The original Cityscapes labels will be grouped into four mutually exclusive project classes. Road and sidewalk will mainly contribute to Walkable. People and vehicles will contribute to Dynamic Obstacle. Fixed objects such as poles and similar structures will contribute to Structural Obstacle. The remaining scene regions will be treated as Non-walkable.

Local fine-tuning: After the public-data baseline, the selected model will be fine-tuned using 300–500 local images. The local set will focus on stairs, curbs, narrow paths, blocked walkways and other campus cases that are not represented well enough in Cityscapes.

Walkability analysis: The segmentation mask will be followed by a simple image-region based analysis. Obstacles that overlap the lower/central part of the expected walking region will be highlighted as a possible obstruction.

5. Experimental and Evaluation Plan

Data Split	**(for local) Training:** 70% **Validation:** 15% **Testing:** 15%
Loss Function	Weighted Cross-Entropy + Dice Loss, with class weights based on training frequencies.
Optimizer	AdamW with a learning-rate scheduler; settings chosen using validation performance.
Evaluation Metrics	IoU, mIoU, Dice Score, Obstacle Recall, Boundary F1, False-Safe Rate, FPS/Latency.
Baseline / Comparison	Cityscapes-only model vs. Cityscapes + campus fine-tuning. BiSeNetV2 vs. Fast-SCNN on accuracy and embedded performance.

6. Expected Outcome / Deliverables

- A trained four-class semantic-segmentation model adapted to campus walking scenes.
- A documented 300–500 image local dataset and clear annotation rules.
- A comparison of BiSeNetV2 and Fast-SCNN using segmentation accuracy and embedded-device performance.
- A TensorRT-optimized inference pipeline running on Jetson Orin Nano.
- A live camera demonstration with walkable-region overlay and obstacle highlighting.

- A report of accuracy, FPS, latency, memory, power and common failure cases.

7. Work Distribution

Sr. No.	Team Member	Major Responsibility
1	Arnav Pangul	Detection pipeline and dataset foundation — public dataset acquisition/preprocessing, custom dataset collection/annotation.
2	Divyansh Gupta	Model Pipeline, Model Training, Segmentation and core reasoning logic.
3	Gauri Thakre	Edge deployment — model conversion (ONNX/TensorRT), Jetson deployment, pipeline integration, latency/FPS benchmarking.
4	Dhruv Runwal	UI, testing and documentation — demo interface, TTS audio alerts, field testing coordination, proposal/report/presentation documentation.

8. Proposed Timeline

Phase	Proposed Activity
Phase 1	**Problem finalization and dataset preparation** —literature survey, dataset acquisition, annotation.
Phase 2	**Preprocessing and baseline** — label remapping and Fast-SCNN/BiSeNetV2 baseline.
Phase 3	**Model development and training** — Model fine-tuning, comparison and hyperparameter improvement
Phase 4	**Edge Deployment** —Jetson Orin Nano deployment, TensorRT optimization and performance testing.
Phase 5	**Evaluation and error analysis** — mIoU, Dice, Recall, FPS/latency, false-positive/negative review.
Phase 6	**Final demonstration, report and presentation** —report writing, demo prep.

9. References

[1] M. Cordts et al., “The Cityscapes Dataset for Semantic Urban Scene Understanding,” Proc. IEEE/CVF CVPR, 2016.

[2] C. Yu et al., “BiSeNet V2: Bilateral Network with Guided Aggregation for Real-time Semantic Segmentation,” arXiv:2004.02147, 2020.

[3] R. P. K. Poudel, S. Liwicki and R. Cipolla, “Fast-SCNN: Fast Semantic Segmentation Network,” arXiv:1902.04502, 2019.

[4] NVIDIA, “NVIDIA TensorRT Documentation,” NVIDIA Developer Documentation.

[5] NVIDIA, “Jetson Orin Nano Developer Kit Gets a Super Boost,” NVIDIA Technical Blog.